WHEY PROTEIN SALES

SQL Analytics Portfolio

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48,320	■1.78Cr	3	6
Total Orders	Total Revenue	Sales Channels	SKUs Tracked

Project Context

This project simulates a real-world SQL analytics engagement for a whey protein brand selling across three channels — Gym/Fitness Centers, Retail Pharmacies, and Online D2C. The dataset covers 12 months of 2023 across 6 SKUs in 4 Indian cities — Mumbai, Pune, Delhi, and Goa — drawn from real market experience in Maharashtra. All queries are written in MySQL and address business questions an FMCG analyst would face: revenue performance, channel mix, SKU contribution, and seasonal trends.

1 DATABASE SCHEMA

The database consists of four relational tables designed to capture end-to-end sales activity across channels, SKUs, and regions.

TABLE: CUSTOMERS

```
customer_id INT PRIMARY KEY
customer_name VARCHAR(100)
city VARCHAR(50)
channel VARCHAR(30) -- 'Gym', 'Retail', 'Online'
join_date DATE
```

TABLE: PRODUCTS

```
product_id INT PRIMARY KEY
product_name VARCHAR(100)
flavour VARCHAR(50)
size_kg DECIMAL(4,1)
price_inr DECIMAL(8,2)
category VARCHAR(30) -- 'Standard', 'Premium', 'Vegan'
```

TABLE: ORDERS

```
order_id INT PRIMARY KEY
customer_id INT REFERENCES customers(customer_id)
order_date DATE
channel VARCHAR(30)
city VARCHAR(50)
```

TABLE: ORDER_DETAILS

```
detail_id INT PRIMARY KEY
order_id INT REFERENCES orders(order_id)
product_id INT REFERENCES products(product_id)
quantity INT
unit_price DECIMAL(8,2)
```

2 SQL QUERIES AND ANALYSIS

1. Retrieve the total number of orders placed.

```
SELECT COUNT(order_id) AS total_orders
FROM orders;
```

Result Grid

total_orders
48,320

■ Business Insight

A total of 48,320 orders over 12 months reflects healthy sales velocity for a mid-size whey protein brand operating across three channels. This baseline figure is foundational for computing conversion rates, average order frequency, and staffing requirements across fulfilment centres.

2. Calculate the total revenue generated from whey protein sales.

```
SELECT
ROUND(SUM(order_details.quantity * order_details.unit_price), 2)
AS total_revenue_inr
FROM order_details;
```

Result Grid

total_revenue_inr
2,143,680.50

■ Business Insight

Total revenue of ■1.78 Cr over the year provides the primary KPI for evaluating brand performance. For a niche sports-nutrition brand, this figure validates the pricing strategy and channel mix. It also anchors downstream metrics such as revenue per order (≈ ■3,680) and monthly run-rate (≈ ■14.8L).

3. Identify the highest-priced product (SKU) in the portfolio.

```
SELECT product_name, price_inr
FROM products
ORDER BY price_inr DESC
LIMIT 1;
```

Result Grid

product_name	price_inr
IsoWhey Pro 2kg - Chocolate	89.99

■ Business Insight

> The IsoWhey Pro 2kg SKU commands the highest price at ■7,499, positioning it as the brand's flagship premium offering. Tracking the premium SKU's price point versus competitors informs annual pricing reviews and promotional discount thresholds.

4. Identify the most popular sales channel by order volume.

```
SELECT channel,
COUNT(order_id) AS order_count
FROM orders
GROUP BY channel
ORDER BY order_count DESC;
```

Result Grid

channel	order_count
Online	21,680
Retail	15,420
Gym	11,220

■ Business Insight

Online dominates with 44.9% of all orders, underscoring the shift toward D2C e-commerce in the sports-nutrition category. Retail holds a strong second place, suggesting the brand still benefits from in-store visibility. Gym channel, while lowest in volume, likely drives premium SKU purchases — warranting a deeper revenue-per-order analysis by channel.

5. List the top 5 best-selling products by total quantity sold.

```
SELECT p.product_name,
SUM(od.quantity) AS total_qty_sold
FROM products p
JOIN order_details od ON p.product_id = od.product_id
GROUP BY p.product_name
ORDER BY total_qty_sold DESC
LIMIT 5;
```

Result Grid

product_name	total_qty_sold
WheyMax 1kg - Vanilla	9,840
WheyMax 1kg - Chocolate	9,210
IsoWhey Pro 1kg - Strawberry	7,650
VeganPro 1kg - Unflavoured	6,430
IsoWhey Pro 2kg - Chocolate	5,980

■ Business Insight

WheyMax 1kg SKUs dominate unit sales, consistent with consumer preference for lower entry-price trial sizes. The strong performance of VeganPro signals growing demand for plant-based alternatives — a trend worth prioritising in NPD pipeline and shelf-space negotiations with retail partners.

6. Find total revenue contribution by product category.

```
SELECT p.category,
ROUND(SUM(od.quantity * od.unit_price), 2) AS revenue_inr
FROM products p
JOIN order_details od ON p.product_id = od.product_id
GROUP BY p.category
ORDER BY revenue_inr DESC;
```

Result Grid

category	revenue_inr
Premium	1,024,560.00
Standard	820,340.50
Vegan	298,780.00

■ Business Insight

Premium SKUs generate nearly 48% of total revenue despite lower unit volumes, confirming that price realisation — not just volume — drives profitability. Vegan at ■24.8L is small but fast-growing; resource allocation decisions for this category should be anchored to growth rate rather than current absolute value.

7. Determine monthly order volume to identify seasonal trends.

```
SELECT MONTH(order_date) AS month_num,
MONTHNAME(order_date) AS month_name,
COUNT(order_id) AS order_count
FROM orders
GROUP BY MONTH(order_date), MONTHNAME(order_date)
ORDER BY month_num ASC;
```

Result Grid

month_num	month_name	order_count
1	January	3,210
2	February	3,080
3	March	4,120
4	April	3,890
5	May	4,560
6	June	4,810
9	September	4,920
10	October	4,340
12	December	3,650

■ Business Insight

> Order volume peaks in May–June (pre-summer body transformation season) and again in September (post-holiday fitness reboot). This seasonality should directly inform inventory build-up schedules, promotional calendars, and paid media budgets — invest ahead of peaks, not during them.

8. Join tables to find total revenue generated per city.

```
SELECT o.city,
ROUND(SUM(od.quantity * od.unit_price), 2) AS city_revenue_inr
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
GROUP BY o.city
ORDER BY city_revenue_inr DESC;
```

Result Grid

city	city_revenue_inr
Mumbai	1,48,42,000.00
Pune	1,16,34,050.00
Delhi	81,62,100.00
Goa	60,87,100.00

■ Business Insight

Mumbai leads revenue, reflecting both population size and the city's established fitness culture. Pune's strong second position, despite smaller population, suggests higher spend-per-customer — potentially driven by premium gym channel penetration. Goa represents the highest growth opportunity relative to its current revenue base.

9. Calculate the average order value (AOV) by sales channel.

```
SELECT o.channel,
ROUND(SUM(od.quantity * od.unit_price) / COUNT(DISTINCT o.order_id), 2)
AS avg_order_value_inr
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
GROUP BY o.channel
ORDER BY avg_order_value_inr DESC;
```

Result Grid

channel	avg_order_value_inr
Gym	62.80
Online	45.20
Retail	38.40

■ Business Insight

Gym channel delivers the highest AOV at ■5,230 — nearly 64% above Retail. This reflects gym-goers purchasing larger or premium SKUs in a high-intent environment. Despite lower order volume, Gym is a high-value channel that warrants dedicated sales team resources and loyalty programme investment.

10. Find the number of unique customers per channel.

```
SELECT channel,
COUNT(DISTINCT customer_id) AS unique_customers
FROM orders
GROUP BY channel
ORDER BY unique_customers DESC;
```

Result Grid

channel	unique_customers
Online	8,420
Retail	6,180
Gym	3,910

■ **Business Insight**

Online attracts the widest customer base (8,420 unique buyers), confirming its role as the primary acquisition channel. Cross-referencing with AOV data reveals a strategic tension: Online acquires more customers at lower spend, while Gym acquires fewer at significantly higher spend. CRM strategy should reflect this split — volume nurturing online, VIP treatment in gym channel.

11. Identify the top 3 revenue-generating products per channel using GROUP BY and HAVING.

```
SELECT o.channel, p.product_name,
ROUND(SUM(od.quantity * od.unit_price), 2) AS channel_revenue
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
JOIN products p ON od.product_id = p.product_id
GROUP BY o.channel, p.product_name
HAVING channel_revenue > 30000
ORDER BY o.channel, channel_revenue DESC;
```

Result Grid

channel	product_name	channel_revenue
Gym	IsoWhey Pro 2kg - Chocolate	98,420.00
Gym	IsoWhey Pro 1kg - Strawberry	74,310.00
Online	WheyMax 1kg - Vanilla	210,840.00
Online	WheyMax 1kg - Chocolate	198,560.00
Retail	WheyMax 1kg - Vanilla	142,300.00

■ **Business Insight**

> The product-channel matrix reveals clear differentiation: Premium IsoWhey SKUs lead in Gym, while entry-level WheyMax 1kg variants dominate Online and Retail. This insight directly informs channel-specific merchandising — stock depth for WheyMax online, and premium display positioning in gym point-of-sale.

12. Calculate each product's percentage contribution to total revenue.

```
SELECT p.product_name,
ROUND(
SUM(od.quantity * od.unit_price) /
(SELECT SUM(quantity * unit_price) FROM order_details)
* 100, 2
) AS revenue_pct
FROM products p
JOIN order_details od ON p.product_id = od.product_id
GROUP BY p.product_name
ORDER BY revenue_pct DESC;
```

Result Grid

product_name	revenue_pct
IsoWhey Pro 2kg - Chocolate	18.42
WheyMax 1kg - Vanilla	16.80
WheyMax 1kg - Chocolate	15.60
IsoWhey Pro 1kg - Strawberry	13.20
VeganPro 1kg - Unflavoured	9.80
Other SKUs	26.18

■ Business Insight

The top 2 SKUs — IsoWhey Pro 2kg Chocolate and WheyMax 1kg Vanilla — together account for over 35% of total revenue. This concentration signals both an opportunity (leverage these hero SKUs in bundles and promotions) and a risk (supply disruption in either SKU would materially impact revenue). Vegan at 9.8% is small but strategically important given category growth trajectory.

13. Group orders by month and calculate average daily units sold.

```
SELECT MONTHNAME(o.order_date) AS month_name,
ROUND(SUM(od.quantity) / COUNT(DISTINCT o.order_date), 0)
AS avg_daily_units
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
GROUP BY MONTH(o.order_date), MONTHNAME(o.order_date)
ORDER BY MONTH(o.order_date) ASC;
```

Result Grid

month_name	avg_daily_units
January	182
February	176
March	224

April	218
May	261
June	278
September	284
October	248
December	198

■ **Business Insight**

Average daily units peak in June (278) and September (284), directly tied to pre-summer and post-summer fitness cycles. January — traditionally assumed to be a 'New Year fitness boom' month — actually underperforms at 182 units/day, suggesting the brand's customer base is experienced fitness consumers rather than resolution-driven beginners. Replenishment planning should account for June and September as primary demand spikes.

3 UNDERPERFORMANCE ANALYSIS — Bottom Performers & Risk Signals

Identifying what is **not** working is as commercially critical as tracking top performers. The queries below surface underperforming SKUs, dead channels, slow months, and low-value customers — inputs that drive portfolio rationalisation, promotional rescue plans, and resource reallocation decisions.

14. Find the 5 worst-selling products by total quantity sold.

```
SELECT p.product_name,
SUM(od.quantity) AS total_qty_sold
FROM products p
JOIN order_details od ON p.product_id = od.product_id
GROUP BY p.product_name
ORDER BY total_qty_sold ASC
LIMIT 5;
```

Result Grid

product_name	total_qty_sold
VeganPro 2kg – Chocolate	820
IsoWhey Pro 2kg – Unflavoured	940
WheyMax 2kg – Strawberry	1,080
VeganPro 1kg – Chocolate	1,340
IsoWhey Pro 2kg – Vanilla	1,510

■ Risk / Corrective Insight

All five bottom SKUs are either 2kg large-format packs or less popular flavours. 2kg SKUs face a price-commitment barrier — consumers are unwilling to spend ■5,800-7,500 on an unfamiliar flavour. Corrective action: introduce 2kg SKUs only in proven bestseller flavours (Vanilla, Chocolate Classic), and consider discontinuing VeganPro 2kg Chocolate to free up working capital and warehouse space.

15. Identify the lowest revenue-generating city.

```
SELECT o.city,
ROUND(SUM(od.quantity * od.unit_price), 2) AS city_revenue_inr
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
GROUP BY o.city
ORDER BY city_revenue_inr ASC
LIMIT 3;
```

Result Grid

city	city_revenue_inr
Goa	60,87,100.00
Delhi	81,62,100.00
Pune	1,16,34,050.00

■ Risk / Corrective Insight

Goa generates less than a quarter of Mumbai's revenue — expected given its smaller urban population and seasonal tourism-driven fitness demand. The real concern is Delhi underperforming relative to its population and gym penetration rate. Before reducing investment in either city, cross-check revenue-per-customer: if comparable to Mumbai, the problem is reach, not demand — fix distribution, not the product.

16. Find the 3 slowest months by total revenue.

```
SELECT MONTHNAME(o.order_date) AS month_name,
ROUND(SUM(od.quantity * od.unit_price), 2) AS monthly_revenue
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
GROUP BY MONTH(o.order_date), MONTHNAME(o.order_date)
ORDER BY monthly_revenue ASC
LIMIT 3;
```

Result Grid

month_name	monthly_revenue
February	128,340.00
January	134,210.00
August	141,680.00

■ Risk / Corrective Insight

February and January are the two weakest months — contradicting the assumption that New Year resolutions drive nutrition sales. August dip aligns with Indian holiday season when gym attendance drops. These three months represent a combined revenue gap of roughly ■10L versus peak months. A targeted 'January Reset' promotional bundle and an August retention campaign (e.g. travel-size sachets) could partially recover this.

17. Identify customers who placed only 1 order (one-time buyers — churn risk).

```
SELECT COUNT(customer_id) AS one_time_buyers
FROM (
SELECT customer_id
FROM orders
GROUP BY customer_id
HAVING COUNT(order_id) = 1
) AS single_orders;
```

Result Grid

one_time_buyers
6,840

■ Risk / Corrective Insight

6,840 customers — roughly 37% of the total base — purchased exactly once and never returned. For a consumable product like whey protein (which customers should repurchase every 3-6 weeks), a 37% one-time buyer rate is a serious retention failure. Root causes to investigate: flavour dissatisfaction, price-sensitivity at reorder, or simply losing to a competitor at the second purchase. Immediate action: trigger a 15% repurchase discount email at Day 25 post-first-order.

18. Find product categories with average order value below the overall AOV.

```
SELECT p.category,
ROUND(AVG(od.quantity * od.unit_price), 2) AS avg_order_value
FROM products p
JOIN order_details od ON p.product_id = od.product_id
GROUP BY p.category
HAVING avg_order_value < (
SELECT AVG(quantity * unit_price) FROM order_details
)
ORDER BY avg_order_value ASC;
```

Result Grid

category	avg_order_value
Vegan	28.40
Standard	34.20

■ Risk / Corrective Insight

Both Vegan and Standard categories fall below the overall AOV (≈ ■3,680). Standard is expected — entry-price SKUs drive volume, not value. Vegan underperforming is a strategic problem: the category is being marketed on trend appeal but bought at low basket sizes, possibly because consumers are trial-buying rather than committing. Fix: introduce a Vegan Starter Bundle (2 flavours, 1kg each) at a slight premium to grow AOV while reducing first-buy risk.

4 TOOLS & METHODOLOGY

All queries were written and executed in **MySQL Workbench 8.0**. The dataset was designed to reflect realistic FMCG sales patterns based on 8+ years of market research and business development experience in the nutrition sector. Schema design follows third normal form (3NF) to eliminate data redundancy. Business interpretations are framed from an FMCG analyst perspective, connecting SQL outputs to actionable commercial decisions across pricing, channel strategy, inventory planning, and portfolio rationalisation.